

BRK (00)		Cycles: 7	Size: 2 *
Implied			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	** Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand.	
2.1		Inc. PC .	
2.2	*** Write to u8(S.L +0) \$0100	Push PC.H onto stack.	
3.1			
3.2	*** Write to u8(S.L -1) \$0100	Push PC.L onto stack.	
4.1		Use Interrupt flags to decide which vector to use. RESET = \$FFFC. NMI = \$FFFA. IRQ/BRK = \$FFFE.	
4.2	*** Write to u8(S.L -2) \$0100	Push P onto stack with B flag if software BRK.	
5.1		Dec. S.L by 3 and set S.H to 1.	
5.2	Read Vector	Store to Address.L.	
6.1		Set I , unset D .	
6.2	Read Vector + 1	Store to Address.H.	
7.1		Enable writes to PC (disabled by NMI/IRQ). Copy Address to PC . Set PB to 0.	
7.2	Read PC:PB	Store as OpCode.	
+X.1			

* Concerning the BRK instruction, you should also note that although its second byte is basically a “don’t care” byte – that is, it can have any value - the BRK (and COP instruction as well) is a two-byte instruction, the second byte sometimes is used as a signature byte to determine the nature of the BRK being executed. When an RTI instruction is executed, control always returns to the second byte past the BRK opcode. — assembly-programming-manual-for-w65c816.pdf

** If NMI, IRQ, or RESET, OpCode is forced to 0 and writes to PC are disabled.

*** If handling a RESET, the writes turn into reads. The databus is populated but ignored.

ORA (01) Cycles: 6		Size: 2
Direct Indexed Indirect (d,x) (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Pointer to Pointer + X + D . Address.L = Pointer.L. Address.H = D .H.
2.2		If D .L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If D .L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer: DB	Store as Operand.
7.1		Perform A = Operand. Set N based off (A .L & \$80) and Z based off A .L.
7.2	Read PC:PB	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

COP (02)	Cycles: 7	Size: 2
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC .
2.2	Write to u8(S .L+0) \$0100	Push PC .H onto stack.
3.1		
3.2	Write to u8(S .L-1) \$0100	Push PC .L onto stack.
4.1		Set Vector to \$FFF4.
4.2	Write to u8(S .L-2) \$0100	Push P onto stack.
5.1		Dec. S .L by 3 and set S .H to 1.
5.2	Read Vector	Store to Address.L.
6.1		Set I , unset D .
6.2	Read Vector + 1	Store to Address.H.
7.1		Copy Address to PC . Set PB to 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (03)		Cycles: 4	Size: 2
Stack Relative (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB	Inc. PC .	
1.2		Store as Pointer.	
2.1		Inc. PC . Set Address to Pointer + (S .L \$100).	
2.2			
3.1			
3.2	Read Address	Store as Operand.	
4.1	Read PC:PB	Perform A = Operand. Set N based off (A .L & \$80) and Z based off A .L.	
4.2		Store as OpCode.	
+X.1			

TSB (04)	Cycles: 5	Size: 2
Direct (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set Z based off (A.L & Operand.L). Perform Operand = A .
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (05)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

ASL (06)	Cycles: 5	Size: 2
Direct (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set C based off (Operand.L & \$80). Perform Operand.L <=< 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (07)	Cycles: 6	Size: 2
Direct Indirect Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

PHP (08)	Cycles: 3	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Write to ((S .L \$0100)+0)	Inc. PC .
1.2		
2.1		Set Operand.L to P with X and M flags set.
2.2		Push Operand.L onto stack.
3.1		Dec. S .L by 1 and set S .H to 1.
3.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (09)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB	Inc. PC .	
1.2		Store as Operand.	
2.1		Inc. PC . Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .	
2.2		Store as OpCode.	
+X.1			

ASL (0A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set C based off (A.L & \$80). Perform A.L <= 1. Set N based off (A.L & \$80) and Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

PHD (0B)	Cycles: 4	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Write to u8(S .L+0) \$0100 Write to ((S .L \$0100)-1) Read PC:PB	Inc. PC .
1.2		
2.1		
2.2		Push D .H onto stack.
3.1		
3.2		Push D .L onto stack.
4.1		
4.2		Store as OpCode.
+X.1		

TSB (0C)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Set Z based off (A .L & Operand.L). Perform Operand = A .
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (0D)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Perform A = Operand. Set N based off (A .L & \$80) and Z based off A .L.
4.2	Read PC:PB	Store as OpCode.
+X.1		

ASL (0E)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Set C based off (Operand.L & \$80). Perform Operand.L <=< 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (0F)	Cycles: 5	Size: 4
Absolute Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Inc. PC .
4.2	Read Address:Bank	Store as Operand.
5.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

BPL (10)	Cycles: 2	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch if N is 0.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

ORA (11)		Cycles: 5	Size: 2
Direct Indirect Indexed (d),y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	Read Address + 1	Store as Pointer.H.	
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).	
5.2			
6.1			
6.2	Read Pointer:Bank	Store as Operand.	
7.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .	
7.2	Read PC:PB	Store as OpCode.	
+X.1			

ORA (12)		Cycles: 5	Size: 2
Direct Indirect (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	Read Address + 1	Store as Pointer.H.	
5.1			
5.2	Read Pointer: DB	Store as Operand.	
6.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .	
6.2	Read PC:PB	Store as OpCode.	
+X.1			

ORA (13)		Cycles: 7	Size: 2
Stack Relative Indirect Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB .	Inc. PC .	
1.2		Store as Pointer.	
2.1		Inc. PC . Set Address to Pointer + (S .L \$100.)	
2.2			
3.1			
3.2		Store as Pointer.	
4.1			
4.2		Store as Pointer.H.	
5.1		Perform Tmp = ui32(Pointer + Y).	
		Pointer = ui16(Tmp).	
5.2		Bank = DB + (Tmp >> 16).	
6.1	Read Pointer:Bank		
6.2		Store as Operand.	
7.1		Perform A = Operand. Set N based off (A .L & \$80) and Z based off A .L.	
7.2	Read PC:PB .	Store as OpCode.	
+X.1			

TRB (14)	Cycles: 5	Size: 2
Direct (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set Z based off (A.L & Operand.L). Perform Operand &= ~ A .
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (15)	Cycles: 4	Size: 2
Direct, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Read PC:PB	Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2		Store as OpCode.
+X.1		

ASL (16)	Cycles: 6	Size: 2
Direct, X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Write to Address	Write Operand.L.
5.2		Set C based off (Operand.L & \$80). Perform Operand.L <=< 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.1		
6.2	Write to Address	Write Operand.L.
7.1	Read PC:PB	
7.2		Store as OpCode.
+X.1		

ORA (17) Cycles: 6		Size: 2
Direct Indirect Indexed Long [d],y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	If D.L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If D.L is not 0, read (Address + 2), otherwise read (Address & \$FF00) ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

CLC (18)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set the C flag to 0.
2.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (19) Cycles: 4		Size: 3
Absolute, Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB .	Store as OpCode.
+X.1		

INC (1A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform A.L += 1, set N based off (A.L & \$80), set Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

TCS (1B)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set S.L to A.L and S.H to 1.
2.2	Read PC:PB	Store as OpCode.
+X.1		

TRB (1C)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Set Z based off (A .L & Operand.L). Perform Operand &= ~ A .
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ORA (1D) Cycles: 4		Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB .	Store as OpCode.
+X.1		

ASL (1E)	Cycles: 7	Size: 3
Absolute, X (Read/Modiy/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\text{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Set C based off (Operand.L & \$80). Perform $\text{Operand.L} \ll= 1$. Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB .	Store as OpCode.
+X.1		

ORA (1F)	Cycles: 5	Size: 4
Absolute Long, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC .
3.2	Read PC:PB .	Stores as Bank.
4.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank += (Tmp >> 16).
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A = Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB .	Store as OpCode.
+X.1		

JSR (20)	Cycles: 6	Size: 3
Absolute		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read PC:PB	Discard.
4.1		Dec. PC .
4.2	Write to u8(S .L+0) \$0100	Push PC .H onto stack.
5.1		
5.2	Write to u8(S .L-1) \$0100	Push PC .L onto stack.
6.1		Dec. S .L by 2 and set S .H to 1. Perform PC = Address.
6.2	Read PC:PB	Store as OpCode.
+X.1		

AND (21)		Cycles: 6	Size: 2
Direct Indexed Indirect (d,x) (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB	Inc. PC .	
1.2		Store as Pointer.	
2.1		Inc. PC . Set Pointer to Pointer + X + D .	
2.2		Address.L = Pointer.L. Address.H = D .H.	
3.1		If D .L is 0, skip the next cycle.	
3.2		Address.H = Pointer.H (fixes high byte of address).	
4.1	Read Address		
4.2		Store as Pointer.	
5.1			
5.2		* If D .L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	
6.1	Read Pointer: DB	Store as Pointer.H.	
6.2		Store as Operand.	
7.1		Perform A = Operand. Set N based off (A .L & \$80) and Z based off A .L.	
7.2		Store as OpCode.	
+X.1			

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

JSL (22)	Cycles: 8	Size: 4
Absolute Long		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Write to u8(S .L+0) \$0100	Push PB onto stack.
4.1		
4.2	Read u8(S .L+0) \$0100	Discard.
5.1		
5.2	Read PC:PB	Stores as Bank.
6.1		
6.2	Write to ((S .L \$0100)-1)	Push PC .H onto stack.
7.1		
7.2	Write to ((S .L \$0100)-2)	Push PC .L onto stack.
8.1		Dec. S .L by 3 and set S .H to 1. Copy Address to PC . Copy Bank to PB .
8.2	Read PC:PB	Store as OpCode.
+X.1		

AND (23)	Cycles: 4	Size: 2
Stack Relative (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.
4.2	Read PC:PB .	Store as OpCode.
+X.1		

BIT (24)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Set Z based off (A.L & Operand.L), set N based off (Operand.L & \$80), and set V based off (Operand.L & \$40).
4.2	Read PC:PB	Store as OpCode.
+X.1		

AND (25)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

ROL (26)		Cycles: 5	Size: 2
Direct (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Write to Address	Write Operand.L.	
5.1		Set C based off (Operand.L & \$80). Perform Operand.L = (Operand.L << 1) C . Set N based off (Operand.L & \$80) and Z based off Operand.L.	
5.2	Write to Address	Write Operand.L.	
6.1			
6.2	Read PC:PB	Store as OpCode.	
+X.1			

AND (27)	Cycles: 6	Size: 2
Direct Indirect Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

PLP (28)	Cycles: 4	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		
2.1		
2.2		
3.1		Inc. S.L by 1 and set S.H to 1.
3.2		Store as Operand.L.
4.1		P = Operand.L, set X and M flags, set X.H and Y.H to 0.
4.2	Read PC:PB	Store as OpCode.
+X.1		

AND (29)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

ROL (2A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set C based off (A.L & \$80). Perform A.L = (A.L << 1) C . Set N based off (A.L & \$80) and Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

PLD (2B)	Cycles: 5	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read ((S.L \$0100)+1)	Inc. PC .
1.2		
2.1		
2.2		
3.1		
3.2		Store as Operand.L.
4.1		
4.2		Store as Operand.H.
5.1		Inc. S.L by 2 and set S.H to 1. Set D to Operand. Set N based off (D.H & \$80). Set Z based off D .
5.2		Store as OpCode.
+X.1		

BIT (2C)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Set Z based off (A .L & Operand.L), set N based off (Operand.L & \$80), and set V based off (Operand.L & \$40).
4.2	Read PC:PB	Store as OpCode.
+X.1		

AND (2D)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.
4.2	Read PC:PB	Store as OpCode.
+X.1		

ROL (2E)		Cycles: 6	Size: 3
Absolute (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC:PB	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address: DB	Store as Operand.	
4.1			
4.2	Write to Address: DB	Write Operand.L.	
5.1		Set C based off (Operand.L & \$80). Perform Operand.L = (Operand.L << 1) C . Set N based off (Operand.L & \$80) and Z based off Operand.L.	
5.2	Write to Address: DB	Write Operand.L.	
6.1			
6.2	Read PC:PB	Store as OpCode.	
+X.1			

AND (2F)		Cycles: 5	Size: 4
Absolute Long (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC:PB	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read PC:PB	Stores as Bank.	
4.1		Inc. PC .	
4.2	Read Address:Bank	Store as Operand.	
5.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .	
5.2	Read PC:PB	Store as OpCode.	
+X.1			

BMI (30)	Cycles: 2	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch if N is 1.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

AND (31) Cycles: 5		Size: 2
Direct Indirect Indexed (d),y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

AND (32)		Cycles: 5	Size: 2
Direct Indirect (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	Read Address + 1	Store as Pointer.H.	
5.1			
5.2	Read Pointer: DB	Store as Operand.	
6.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .	
6.2	Read PC:PB	Store as OpCode.	
+X.1			

AND (33)		Cycles: 7	Size: 2
Stack Relative Indirect Indexed (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB .	Inc. PC .	
1.2		Store as Pointer.	
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).	
2.2			
3.1			
3.2	Read Address	Store as Pointer.	
4.1	Read Address + 1	Store as Pointer.H.	
4.2		Perform Tmp = ui32(Pointer + Y).	
5.1		Pointer = ui16(Tmp).	
		Bank = DB + (Tmp >> 16).	
5.2			
6.1	Read Pointer:Bank	Store as Operand.	
6.2		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.	
7.1			
7.2	Read PC:PB .	Store as OpCode.	
+X.1			

BIT (34)	Cycles: 4	Size: 2
Direct, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB .	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Read PC:PB .	Set Z based off (A.L & Operand.L), set N based off (Operand.L & \$80), and set V based off (Operand.L & \$40).
5.2		Store as OpCode.
+X.1		

AND (35)		Cycles: 4	Size: 2
Direct, X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB	Inc. PC .	
1.2		Store as Operand.	
2.1		Inc. PC .	
2.2			
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.	
3.2			
4.1			
4.2	Read Address	Store as Operand.	
5.1	Read PC:PB	Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .	
5.2		Store as OpCode.	
+X.1			

ROL (36)	Cycles: 6	Size: 2
Direct Indexed Indirect (d,x) (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Write to Address	Write Operand.L.
5.2		Set C based off (Operand.L & \$80). Perform Operand.L = (Operand.L << 1) C . Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.1		
6.2	Write to Address	Write Operand.L.
7.1	Read PC:PB	
7.2		Store as OpCode.
+X.1		

AND (37)		Cycles: 6	Size: 2
Direct Indirect Indexed Long [d],y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	If D.L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.	
5.1			
5.2	If D.L is not 0, read (Address + 2), otherwise read (Address & \$FF00) ui8(Address.L + 2)	Store as Bank.	
6.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank += (Tmp >> 16).	
6.2	Read Pointer:Bank	Store as Operand.	
7.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .	
7.2	Read PC:PB	Store as OpCode.	
+X.1			

SEC (38)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set the C flag to 1.
2.2	Read PC:PB	Store as OpCode.
+X.1		

AND (39)		Cycles: 4	Size: 3
Absolute, Y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC:PB	Store as Pointer.H.	
3.1		Inc. PC . Perform Orig = Pointer.	
		Tmp = ui32(Pointer + Y).	
		Pointer = ui16(Tmp).	
		If Orig.H == Pointer.H, skip 2 half-cycles.	
		Bank = DB + (Tmp >> 16).	
3.2			
4.1			
4.2	Read Pointer:Bank	Store as Operand.	
5.1		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.	
5.2	Read PC:PB	Store as OpCode.	
+X.1			

DEC (3A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform A.L -= 1, set N based off (A.L & \$80), set Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

TSC (3B)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set A to (S .L \$0100). Set N based off (A .H & \$80) and Z based off A .
2.2	Read PC:PB	Store as OpCode.
+X.1		

BIT (3C)	Cycles: 4	Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set Z based off (A .L & Operand.L), set N based off (Operand.L & \$80), and set V based off (Operand.L & \$40).
5.2	Read PC:PB	Store as OpCode.
+X.1		

AND (3D) Cycles: 4		Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A &= Operand. Set N based off (A .L & \$80) and Z based off A .L.
5.2	Read PC:PB	Store as OpCode.
+X.1		

ROL (3E)	Cycles: 7	Size: 3
Absolute, X (Read/Modiy/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\text{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Set C based off (Operand.L & \$80). Perform $\text{Operand.L} = (\text{Operand.L} \ll 1) \mid \text{C}$. Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB .	Store as OpCode.
+X.1		

AND (3F)		Cycles: 5	Size: 4
Absolute Long, X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB .	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC:PB .	Store as Pointer.H.	
3.1		Inc. PC .	
3.2	Read PC:PB .	Stores as Bank.	
4.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$.	
		Pointer = $\text{ui16}(\text{Tmp})$.	
		Bank += (Tmp >> 16).	
4.2	Read Pointer:Bank	Store as Operand.	
5.1		Perform A &= Operand. Set N based off (A.L & \$80) and Z based off A.L .	
5.2	Read PC:PB .	Store as OpCode.	
+X.1			

RTI (40)	Cycles: 6	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2		
2.1		
2.2		
3.1		Inc. S.L by 1 and set S.H to 1.
3.2	Read ((S.L \$0100)+0)	Store as Operand.L.
4.1		Set P to ((Operand.L & ~\$30) (P & \$30)).
4.2	Read ((S.L \$0100)+1)	Store as Operand.L.
5.1		
5.2	Read ((S.L \$0100)+2)	Store as Operand.H.
6.1		Inc. S.L by 2 and set S.H to 1. Set PC to Operand.
6.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (41)		Cycles: 6	Size: 2
Direct Indexed Indirect (d,x) (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB	Inc. PC .	
1.2		Store as Pointer.	
2.1		Inc. PC . Set Pointer to Pointer + X + D .	
2.2		Address.L = Pointer.L. Address.H = D .H.	
3.1		If D .L is 0, skip the next cycle.	
3.2		Address.H = Pointer.H (fixes high byte of address).	
4.1	Read Address		
4.2		Store as Pointer.	
5.1			
5.2		* If D .L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	
6.1	Read Pointer: DB		
6.2		Store as Operand.	
7.1		Perform A ^= Operand. Set N based off (A .L & \$80) and Z based off A .L.	
7.2		Store as OpCode.	
+X.1			

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

WDM (42) Cycles: 2		Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC .
2.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (43)	Cycles: 4	Size: 2
Stack Relative (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1		Perform A ^= Operand. Set N based off (A .L & \$80) and Z based off A .L.
4.2	Read PC:PB	Store as OpCode.
+X.1		

MVP (44)	Cycles: 7	Size: 3
Block Move (Positive)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Set DB to Operand.L.
2.2	Read PC:PB	Stores as Bank.
3.1		Inc. PC .
3.2	Read X:Bank	Store as Operand.L.
4.1		
4.2	Write Y:DB	Write Operand.L.
5.1		Perform A -= 1. If X flag is set, perform X.L -= 1 and Y.L -= 1, otherwise perform X -= 1 and Y -= 1. If A is not \$FFFF, perform PC -= 3.
5.2		
6.1		
6.2		
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (45)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

LSR (46)	Cycles: 5	Size: 2
Direct (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Set C based off (Operand.L & \$01). Perform Operand.L >>= 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (47)	Cycles: 6	Size: 2
Direct Indirect Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

PHA (48)	Cycles: 3	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Write to ((S.L \$0100)+0)	Inc. PC .
1.2		
2.1		
2.2		Push A.L onto stack.
3.1		Dec. S.L by 1 and set S.H to 1.
3.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (49)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Perform A ^= Operand. Set N based off (A .L & \$80) and Z based off A .L.
2.2	Read PC:PB	Store as OpCode.
+X.1		

LSR (4A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set C based off (A.L & \$01). Perform A.L >>= 1. Set N based off (A.L & \$80) and Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

PHK (4B)	Cycles: 3	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Write to ((S.L \$0100)+0)	Inc. PC .
1.2		
2.1		
2.2		Push PB onto stack.
3.1		Dec. S.L by 1 and set S.H to 1.
3.2	Read PC:PB	Store as OpCode.
+X.1		

JMP (4C)		Cycles: 3	Size: 3
Absolute			
Cycle	R/W		Desc.
-X.2	Read PC:PB		Store as OpCode.
1.1			Inc. PC .
1.2	Read PC:PB		Store as Address.
2.1			Inc. PC .
2.2	Read PC:PB		Store as Address.H.
3.1			Set PC to Address.
3.2	Read PC:PB		Store as OpCode.
+X.1			

EOR (4D)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

LSR (4E)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Set C based off (Operand.L & \$01). Perform Operand.L >>= 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (4F)	Cycles: 5	Size: 4
Absolute Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Inc. PC .
4.2	Read Address:Bank	Store as Operand.
5.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

BVC (50)	Cycles: 2	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch if V is 0.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

EOR (51)	Cycles: 5	Size: 2
Direct Indirect Indexed (d),y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (52)	Cycles: 5	Size: 2
Direct Indirect (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer: DB	Store as Operand.
6.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
6.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (53)	Cycles: 7	Size: 2
Stack Relative Indirect Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Perform A ^= Operand. Set N based off (A .L & \$80) and Z based off A .L.
7.2	Read PC:PB	Store as OpCode.
+X.1		

MVN (54)	Cycles: 7	Size: 3
Block Move (Negative)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Set DB to Operand.L.
2.2	Read PC:PB	Stores as Bank.
3.1		Inc. PC .
3.2	Read X:Bank	Store as Operand.L.
4.1		
4.2	Write Y:DB	Write Operand.L.
5.1		Perform A -= 1. If X flag is set, perform X.L += 1 and Y.L += 1, otherwise perform X += 1 and Y += 1. If A is not \$FFFF, perform PC -= 3.
5.2		
6.1		
6.2		
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (55)	Cycles: 4	Size: 2
Direct, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Read PC:PB	Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2		Store as OpCode.
+X.1		

LSR (56)	Cycles: 6	Size: 2
Direct, X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Write to Address	Write Operand.L.
5.2		Set C based off (Operand.L & \$01). Perform Operand.L >>= 1. Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.1		
6.2	Write to Address	Write Operand.L.
7.1	Read PC:PB	
7.2		Store as OpCode.
+X.1		

EOR (57)		Cycles: 6	Size: 2
Direct Indirect Indexed Long [d],y (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	If D.L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.	
5.1			
5.2	If D.L is not 0, read (Address + 2), otherwise read (Address & \$FF00) ui8(Address.L + 2)	Store as Bank.	
6.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank += (Tmp >> 16).	
6.2	Read Pointer:Bank	Store as Operand.	
7.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .	
7.2	Read PC:PB	Store as OpCode.	
+X.1			

CLI (58)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set the I flag to 0.
2.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (59)	Cycles: 4	Size: 3
Absolute, Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

PHY (5A)	Cycles: 3	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Write to ((S.L \$0100)+0)	Inc. PC .
1.2		
2.1		
2.2		Push Y.L onto stack.
3.1		Dec. S.L by 1 and set S.H to 1.
3.2	Read PC:PB	Store as OpCode.
+X.1		

TCD (5B)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set D to A . Set N based off (D .H & \$80) and Z based off D .
2.2	Read PC:PB	Store as OpCode.
+X.1		

JML (5C)	Cycles: 4	Size: 4
Absolute Long		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Copy Address to PC . Copy Bank to PB .
4.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (5D) Cycles: 4		Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB .	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB .	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Perform A ^= Operand. Set N based off (A.L & \$80) and Z based off A.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

LSR (5E)	Cycles: 7	Size: 3
Absolute, X (Read/Modiy/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\mathbf{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Set C based off (Operand.L & \$01). Perform $\text{Operand.L} \gg= 1$. Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

EOR (5F)		Cycles: 5	Size: 4
Absolute Long, X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC:PB	Store as Pointer.H.	
3.1		Inc. PC .	
3.2	Read PC:PB	Stores as Bank.	
4.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$.	
		Pointer = $\text{ui16}(\text{Tmp})$.	
		Bank += (Tmp >> 16).	
4.2	Read Pointer:Bank	Store as Operand.	
5.1		Perform $\text{A} \wedge = \text{Operand}$. Set N based off (A.L & \$80) and Z based off A.L .	
5.2	Read PC:PB	Store as OpCode.	
+X.1			

RTS (60)	Cycles: 6	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read ((S.L \$0100)+0)	Inc. PC .
1.2		
2.1		
2.2		
3.1		Inc. S.L by 1 and set S.H to 1.
3.2		Store as Operand.L.
4.1		
4.2		Store as Operand.H.
5.1		
5.2		
6.1		Inc. S.L by 1 and set S.H to 1. Set PC to (Operand + 1).
6.2		Store as OpCode.
+X.1		

ADC (61) Cycles: 6		Size: 2
Direct Indexed Indirect (d,x) (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Pointer to Pointer + X + D . Address.L = Pointer.L. Address.H = D .H.
2.2		If D .L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If D .L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer: DB	Store as Operand.
7.1		If D flag == 0: Result = ui16(A .L) + ui16(Operand) + C . V flag = (~ui16(A .L) ^ ui16(Operand)) & (ui16(A .L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A .L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A .L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A .L >> 4) ^ (Operand >> 4)) & ((A .L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

PER (62)	Cycles: 6	Size: 3
Program Counter Relative Long		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Operand.H.
3.1		Inc. PC . Set Operand to Operand + PC .
3.2		
4.1		
4.2	Write to ((S .L \$0100)+0)	Push Operand.H onto stack.
5.1		
5.2	Write to ((S .L \$0100)-1)	Push Operand.L onto stack.
6.1		Dec. S .L by 2 and set S .H to 1.
6.2	Read PC:PB	Store as OpCode.
+X.1		

ADC (63)		Cycles: 4	Size: 2
Stack Relative (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB	Inc. PC .	
1.2		Store as Pointer.	
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).	
2.2			
3.1			
3.2	Read Address	Store as Operand.	
4.1		If D flag == 0: Result = ui16(A .L) + ui16(Operand) + C . V flag = (~(ui16(A .L) ^ ui16(Operand)) & (ui16(A .L) ^ ui8(Result)) & \$80) != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A .L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A .L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A .L >> 4) ^ (Operand >> 4)) & ((A .L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.	
4.2		Store as OpCode.	
+X.1			

STZ (64)	Cycles: 3	Size: 2
Direct (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Write to Address	Write 0.
4.1		
4.2	Read PC:PB	Store as OpCode.
+X.1		

ADC (65)		Cycles: 3	Size: 2
Direct (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Operand.	
4.1		If D flag == 0: Result = ui16(A.L) + ui16(Operand) + C . V flag = (~ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A.L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A.L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.	
4.2	Read PC:PB	Store as OpCode.	
+X.1			

ROR (66)		Cycles: 5	Size: 2
Direct (Read/Modify/Write)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Operand.	
4.1			
4.2	Write to Address	Write Operand.L.	
5.1		Set C based off (Operand.L & \$01). Perform Operand.L = (Operand.L >> 1) (C << 7). Set N based off (Operand.L & \$80) and Z based off Operand.L.	
5.2	Write to Address	Write Operand.L.	
6.1			
6.2	Read PC:PB	Store as OpCode.	
+X.1			

ADC (67) Cycles: 6		Size: 2
Direct Indirect Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		If D flag == 0: $Result = ui16(A.L) + ui16(Operand) + C.$ $V\text{ flag} = (\sim(ui16(A.L) \wedge ui16(Operand)) \& (ui16(A.L) \wedge ui8(Result)) \& \$80) \neq 0.$ $C\text{ flag} = Result > \$FF.$ $Result = ui8(Result). Z\text{ flag} = Result == \$00. N\text{ flag} = (Result \& \$80) \neq 0.$ If D flag == 1: $Lo = ui16(A.L \& 0x0F) + ui16(Operand \& 0x0F) + C;$ if $(Lo > 9) Lo += 6.$ $CarryToHi = Lo > \$0F ? 1 : 0.$ $HiSum = ui16(A.L >> 4) + ui16(Operand >> 4) + CarryToHi.$ $V\text{ flag} = (\sim((A.L >> 4) \wedge (Operand >> 4)) \& ((A.L >> 4) \wedge HiSum) \& \$08) \neq 0.$ if $(HiSum > 9) HiSum += 6.$ $C\text{ flag} = HiSum > \$0F.$ $Result = ui8(((HiSum \& 0x0F) << 4) (Lo \& \$0F)).$ $Z\text{ flag} = Result == \$00. N\text{ flag} = (Result \& \$80) \neq 0.$
7.2	Read PC:PB	Store as OpCode.
+X.1		

PLA (68)	Cycles: 4	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		
2.1		
2.2		
3.1		Inc. S.L by 1 and set S.H to 1.
3.2		Store as Operand.L.
4.1		Set A.L to Operand.L. Set N based off (A.L & \$80) and Z based off A.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

ADC (69)		Cycles: 2	Size: 2
Immediate			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand.	
2.1		If D flag == 0: Result = ui16(A.L) + ui16(Operand) + C . V flag = (~ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A.L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A.L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = ~(A.L >> 4) ^ (Operand >> 4) & ((A.L >> 4) ^ HiSum) & \$08 != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.	
2.2	Read PC:PB	Store as OpCode.	
+X.1			

ROR (6A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set C based off (A.L & \$01). Perform A.L = (A.L >> 1) (C << 7). Set N based off (A.L & \$80) and Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

RTL (6B)	Cycles: 6	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2		
2.1		
2.2		
3.1		
3.2	Read u8(S .L+1) \$0100	Store as Operand.L.
4.1		
4.2	Read u8(S .L+2) \$0100	Store as Operand.H.
5.1		
5.2	Read ((S .L \$0100)+3)	Store as PB .
6.1		Inc. S .L by 3 and set S .H to 1. Set PC to (Operand + 1). Set PB to Bank.
6.2	Read PC:PB	Store as OpCode.
+X.1		

JMP (6C) Cycles: 5 Size: 3		
Absolute Indirect (Jump)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC .
3.2	Read Pointer	Store as Address.
4.1		
4.2	Read Pointer + 1	Store as Address.H.
5.1		Set PC to Address.
5.2	Read PC:PB	Store as OpCode.
+X.1		

ADC (6D)		Cycles: 4	Size: 3
Absolute (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC:PB	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read Address: DB	Store as Operand.	
4.1		If D flag == 0: Result = ui16(A.L) + ui16(Operand) + C . V flag = (~ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A.L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A.L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.	
4.2	Read PC:PB	Store as OpCode.	
+X.1			

ROR (6E)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Set C based off (Operand.L & \$01). Perform Operand.L = (Operand.L >> 1) (C << 7). Set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

ADC (6F)		Cycles: 5	Size: 4
Absolute Long (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Address.	
2.1		Inc. PC .	
2.2	Read PC:PB	Store as Address.H.	
3.1		Inc. PC .	
3.2	Read PC:PB	Stores as Bank.	
4.1		Inc. PC .	
4.2	Read Address:Bank	Store as Operand.	
5.1		If D flag == 0: Result = ui16(A.L) + ui16(Operand) + C . V flag = (~ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A.L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A.L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.	
5.2	Read PC:PB	Store as OpCode.	
+X.1			

BVS (70)	Cycles: 2	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch if V is 1.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

ADC (71) Cycles: 5		Size: 2
Direct Indirect Indexed (d),y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		If D flag == 0: Result = ui16(A.L) + ui16(Operand) + C . V flag = (~ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A.L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A.L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

ADC (72)		Cycles: 5	Size: 2
Direct Indirect (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	Read Address + 1	Store as Pointer.H.	
5.1			
5.2	Read Pointer: DB	Store as Operand.	
6.1		If D flag == 0: $\text{Result} = \text{ui16}(\mathbf{A.L}) + \text{ui16}(\text{Operand}) + \mathbf{C}.$ $\mathbf{V} \text{ flag} = (\sim(\text{ui16}(\mathbf{A.L}) \wedge \text{ui16}(\text{Operand})) \& (\text{ui16}(\mathbf{A.L}) \wedge \text{ui8}(\text{Result})) \& \$80) \neq 0.$ $\mathbf{C} \text{ flag} = \text{Result} > \$\text{FF}.$ $\text{Result} = \text{ui8}(\text{Result}). \mathbf{Z} \text{ flag} = \text{Result} == \$00. \mathbf{N} \text{ flag} = (\text{Result} \& \$80) \neq 0.$ If D flag == 1: $\text{Lo} = \text{ui16}(\mathbf{A.L} \& 0x0\text{F}) + \text{ui16}(\text{Operand} \& 0x0\text{F}) + \mathbf{C};$ if $(\text{Lo} > 9) \text{ Lo} += 6.$ $\text{CarryToHi} = \text{Lo} > \$0\text{F} ? 1 : 0.$ $\text{HiSum} = \text{ui16}(\mathbf{A.L} \gg 4) + \text{ui16}(\text{Operand} \gg 4) + \text{CarryToHi}.$ $\mathbf{V} \text{ flag} = (\sim((\mathbf{A.L} \gg 4) \wedge (\text{Operand} \gg 4)) \& ((\mathbf{A.L} \gg 4) \wedge \text{HiSum}) \& \$08) \neq 0.$ if $(\text{HiSum} > 9) \text{ HiSum} += 6.$ $\mathbf{C} \text{ flag} = \text{HiSum} > \$0\text{F}.$ $\text{Result} = \text{ui8}(((\text{HiSum} \& 0x0\text{F}) \ll 4) \mid (\text{Lo} \& \$0\text{F})).$ $\mathbf{Z} \text{ flag} = \text{Result} == \$00. \mathbf{N} \text{ flag} = (\text{Result} \& \$80) \neq 0.$	
6.2	Read PC:PB	Store as OpCode.	
+X.1			

ADC (73) Cycles: 7		Size: 2
Stack Relative Indirect Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S.L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		If D flag == 0: Result = ui16(A.L) + ui16(Operand) + C . V flag = (~ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A.L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A.L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

STZ (74)	Cycles: 4	Size: 2
Direct, X (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2	Write to Address	
4.1		
4.2		Write 0.
5.1	Read PC:PB	
5.2		Store as OpCode.
+X.1		

ADC (75) Cycles: 4		Size: 2
Direct, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		If D flag == 0: Result = ui16(A.L) + ui16(Operand) + C . V flag = (~ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A.L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A.L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
5.2	Read PC:PB	Store as OpCode.
+X.1		

ROR (76)	Cycles: 6	Size: 2
Direct, X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		
5.2	Write to Address	Write Operand.L.
6.1		Set C based off (Operand.L & \$01). Perform Operand.L = (Operand.L >> 1) (C << 7). Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.2	Write to Address	Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

ADC (77) Cycles: 6		Size: 2
Direct Indirect Indexed Long [d],y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	If D.L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If D.L is not 0, read (Address + 2), otherwise read (Address & \$FF00) ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.
7.1		If D flag == 0: Result = ui16(A.L) + ui16(Operand) + C . V flag = (~ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A.L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A.L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

SEI (78)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set the I flag to 1.
2.2	Read PC:PB	Store as OpCode.
+X.1		

ADC (79) Cycles: 4		Size: 3
Absolute, Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		If D flag == 0: Result = ui16(A.L) + ui16(Operand) + C . V flag = (~ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A.L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A.L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
5.2	Read PC:PB	Store as OpCode.
+X.1		

PLY (7A)	Cycles: 4	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2		
2.1		
2.2		
3.1		Inc. S.L by 1 and set S.H to 1.
3.2		Store as Operand.L.
4.1	Set Y.L to Operand.L. Set N based off (Y.L & \$80). Set Z based off Y.L .	
4.2	Read PC:PB	Store as OpCode.
+X.1		

TDC (7B)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set A to D . Set N based off (A .H & \$80) and Z based off A .
2.2	Read PC:PB	Store as OpCode.
+X.1		

JMP (7C) Cycles: 6		Size: 3
Absolute Indexed Indirect (a,x) (Jump		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Perform Pointer += X . Bank = PB .
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Address.L.
5.1		
5.2	Read Pointer + 1:Bank	Store as Address.H.
6.1		Set PC to Address.
6.2	Read PC:PB	Store as OpCode.
+X.1		

ADC (7D) Cycles: 4		Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC. Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		If D flag == 0: Result = ui16(A.L) + ui16(Operand) + C . V flag = (~ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80 != 0. C flag = Result > \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = ui16(A.L & 0x0F) + ui16(Operand & 0x0F) + C ; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = ui16(A.L >> 4) + ui16(Operand >> 4) + CarryToHi. V flag = (~((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
5.2	Read PC:PB	Store as OpCode.
+X.1		

ROR (7E)	Cycles: 7	Size: 3
Absolute, X (Read/Modiy/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\text{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Set C based off (Operand.L & \$01). Perform $\text{Operand.L} = (\text{Operand.L} \gg 1) \mid (\text{C} \ll 7)$. Set N based off (Operand.L & \$80) and Z based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

ADC (7F)		Cycles: 5	Size: 4
Absolute Long, X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1	Read PC:PB	Inc. PC .	
1.2		Store as Pointer.	
2.1	Read PC:PB	Inc. PC .	
2.2		Store as Pointer.H.	
3.1	Read PC:PB	Inc. PC .	
3.2		Stores as Bank.	
4.1	Read Pointer:Bank	Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$.	
4.2		Pointer = $\text{ui16}(\text{Tmp})$. Bank += $(\text{Tmp} \gg 16)$.	
5.1	Read PC:PB	Store as Operand.	
5.2		If D flag == 0: Result = $\text{ui16}(\text{A.L}) + \text{ui16}(\text{Operand}) + \text{C}$. V flag = $(\sim(\text{ui16}(\text{A.L}) \wedge \text{ui16}(\text{Operand})) \& (\text{ui16}(\text{A.L}) \wedge \text{ui8}(\text{Result})) \& \$80) \neq 0$. C flag = Result > \$FF. Result = $\text{ui8}(\text{Result})$. Z flag = Result == \$00. N flag = $(\text{Result} \& \$80) \neq 0$. If D flag == 1: Lo = $\text{ui16}(\text{A.L} \& 0x0F) + \text{ui16}(\text{Operand} \& 0x0F) + \text{C}$; if (Lo > 9) Lo += 6. CarryToHi = Lo > \$0F ? 1 : 0. HiSum = $\text{ui16}(\text{A.L} \gg 4) + \text{ui16}(\text{Operand} \gg 4) + \text{CarryToHi}$. V flag = $(\sim((\text{A.L} \gg 4) \wedge (\text{Operand} \gg 4)) \& ((\text{A.L} \gg 4) \wedge \text{HiSum}) \& \$08) \neq 0$. if (HiSum > 9) HiSum += 6. C flag = HiSum > \$0F. Result = $\text{ui8}(((\text{HiSum} \& 0x0F) \ll 4) \mid (\text{Lo} \& \$0F))$. Z flag = Result == \$00. N flag = $(\text{Result} \& \$80) \neq 0$.	
+X.1		Store as OpCode.	

BRA (80)	Cycles: 3	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch always.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

STA (81)	Cycles: 6	Size: 2
Direct Indexed Indirect (d,x) (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Pointer to Pointer + X + D . Address.L = Pointer.L. Address.H = D .H.
2.2		If D .L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If D .L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
6.1		Set Operand to A.
6.2	Write to Pointer: DB	Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

BRL (82)	Cycles: 4	Size: 3
Program Counter Relative Long		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Operand.H.
3.1		Inc. PC .
3.2		
4.1		Perform PC += Operand.
4.2	Read PC:PB	Store as OpCode.
+X.1		

STA (83)	Cycles: 4	Size: 2
Stack Relative (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		Set Operand to A .
3.2	Write to Address	Write Operand.L.
4.1		
4.2	Read PC:PB	Store as OpCode.
+X.1		

STY (84)	Cycles: 3	Size: 2
Direct (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Write to Address	Write Y.L .
4.1		
4.2	Read PC:PB	Store as OpCode.
+X.1		

STA (85)	Cycles: 3	Size: 2
Direct (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Write to Address	Write A.L .
4.1		
4.2	Read PC:PB	Store as OpCode.
+X.1		

STX (86)	Cycles: 3	Size: 2
Direct (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Write to Address	Write X.L .
4.1		
4.2	Read PC:PB	Store as OpCode.
+X.1		

STA (87)	Cycles: 6	Size: 2
Direct Indirect Long (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		Set Operand to A .
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

DEY (88)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform Y.L -= 1, set N based off (Y.L & \$80), set Z based off Y.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

BIT (89)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Set Z based off (A.L & Operand.L).
2.2	Read PC:PB	Store as OpCode.
+X.1		

TXA (8A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform A.L = X.L , set N based off (A.L & \$80), set Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

PHB (8B)		Cycles: 3	Size: 1
Implied			
Cycle	R/W		Desc.
-X.2	Read PC:PB		Store as OpCode.
1.1			Inc. PC .
1.2			
2.1			
2.2	Write to ((S.L \$0100)+0)		Push DB onto stack.
3.1			Dec. S.L by 1 and set S.H to 1.
3.2	Read PC:PB		Store as OpCode.
+X.1			

STY (8C)	Cycles: 4	Size: 3
Absolute (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc PC . Set Operand to Y .
3.2	Write to Address: DB	Write Operand.L.
4.1		
4.2	Read PC:PB	Store as OpCode.
+X.1		

STA (8D)	Cycles: 4	Size: 3
Absolute (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc PC . Set Operand to A .
3.2	Write to Address: DB	Write Operand.L.
4.1		
4.2	Read PC:PB	Store as OpCode.
+X.1		

STX (8E)	Cycles: 4	Size: 3
Absolute (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc PC . Set Operand to X .
3.2	Write to Address: DB	Write Operand.L.
4.1		
4.2	Read PC:PB	Store as OpCode.
+X.1		

STA (8F)	Cycles: 5	Size: 4
Absolute Long (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Inc PC . Set Operand to A .
4.2	Write to Address:Bank	Write Operand.L.
5.1		
5.2	Read PC:PB	Store as OpCode.
+X.1		

BCC (90)	Cycles: 2	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch if C is 0.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

STA (91)	Cycles: 6	Size: 2
Direct Indirect Indexed (d),y (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{Y})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\text{DB} + (\text{Tmp} \gg 16)$.
5.2		
6.1		Set Operand to A .
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

STA (92)	Cycles: 5	Size: 2
Direct Indirect (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Set Operand to A .
5.2	Write to Pointer: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

STA (93)	Cycles: 7	Size: 2
Stack Relative Indirect Indexed (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank = DB + (Tmp >> 16).
5.2		
6.1		Set Operand to A .
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

STY (94)	Cycles: 4	Size: 2
Direct, X (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Write to Address	Write Y.L .
5.1	Read PC:PB	
5.2		Store as OpCode.
+X.1		

STA (95)	Cycles: 4	Size: 2
Direct, X (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Write to Address	Write A.L .
5.1	Read PC:PB	Store as OpCode.
5.2		
+X.1		

STX (96)	Cycles: 4	Size: 2
Direct, Y (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + Y.L + D.L , Address.H = D.H , otherwise Address = Operand + Y + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Write to Address	Write X.L .
5.1		
5.2	Read PC:PB	Store as OpCode.
+X.1		

STA (97)	Cycles: 6	Size: 2
Direct Indirect Indexed Long [d],y (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	If D.L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If D.L is not 0, read (Address + 2), otherwise read (Address & \$FF00) ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Write to Pointer:Bank	Write A.L .
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

TYA (98)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform A.L = Y.L , set N based off (A.L & \$80), set Z based off A.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

STA (99)	Cycles: 5	Size: 3
Absolute, Y (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{Y})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\text{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		Set Operand to A .
4.2	Write to Pointer:Bank	Write Operand.L.
5.1		
5.2	Read PC:PB	Store as OpCode.
+X.1		

TXS (9A)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set S.L to X.L and S.H to 1
2.2	Read PC:PB	Store as OpCode.
+X.1		

TXY (9B)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform Y.L = X.L , set N based off (Y.L & \$80), set Z based off Y.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

STZ (9C)	Cycles: 4	Size: 3
Absolute (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc PC . Set Operand to 0.
3.2	Write to Address: DB	Write Operand.L.
4.1		
4.2	Read PC:PB	Store as OpCode.
+X.1		

STA (9D)	Cycles: 5	Size: 3
Absolute, X (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\mathbf{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		Set Operand to A .
4.2	Write to Pointer:Bank	Write Operand.L.
5.1		
5.2	Read PC:PB	Store as OpCode.
+X.1		

STZ (9E)	Cycles: 5	Size: 3
Absolute, X (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\mathbf{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		Set Operand to 0.
4.2	Write to Pointer:Bank	Write Operand.L.
5.1		
5.2	Read PC:PB	Store as OpCode.
+X.1		

STA (9F)	Cycles: 5	Size: 4
Absolute Long, X (Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank += (Tmp >> 16).
4.2	Write to Pointer:Bank	Write A.L .
5.1		
5.2	Read PC:PB	Store as OpCode.
+X.1		

LDY (A0)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Set Y.L to Operand.L. Set N based off (Y.L & \$80). Set Z based off Y.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (A1)	Cycles: 6	Size: 2
Direct Indexed Indirect (d,x) (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Pointer to Pointer + X + D . Address.L = Pointer.L. Address.H = D .H.
2.2		If D .L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If D .L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer: DB	Store as Operand.
7.1		Set A .L to Operand.L. Set N based off (A .L & \$80). Set Z based off A .L.
7.2	Read PC:PB	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

LDX (A2)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Set X.L to Operand.L. Set N based off (X.L & \$80). Set Z based off X.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (A3)	Cycles: 4	Size: 2
Stack Relative (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1	Read PC:PB	Set A .L to Operand.L. Set N based off (A .L & \$80). Set Z based off A .L.
4.2		Store as OpCode.
+X.1		

LDY (A4)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Set Y.L to Operand.L. Set N based off (Y.L & \$80). Set Z based off Y.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (A5)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Set A.L to Operand.L. Set N based off (A.L & \$80). Set Z based off A.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

LDX (A6)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Set X.L to Operand.L. Set N based off (X.L & \$80). Set Z based off X.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (A7)	Cycles: 6	Size: 2
Direct Indirect Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set A.L to Operand.L. Set N based off (A.L & \$80). Set Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

TAY (A8)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform Y.L = A.L , set N based off (Y.L & \$80), set Z based off Y.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (A9)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC . Set A .L to Operand.L. Set N based off (A .L & \$80). Set Z based off A .L.
2.2	Read PC:PB	Store as OpCode.
+X.1		

TAX (AA)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform X.L = A.L , set N based off (X.L & \$80), set Z based off X.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

PLB (AB)	Cycles: 4	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		
2.1		
2.2		
3.1		Inc. S.L by 1 and set S.H to 1.
3.2		Store as Bank.
4.1		Set DB to Bank, set N based off (DB & \$80), and set Z based off DB .
4.2		Store as OpCode.
+X.1		

LDY (AC)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Set Y .L to Operand.L. Set N based off (Y .L & \$80). Set Z based off Y .L.
4.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (AD)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Set A.L to Operand.L. Set N based off (A.L & \$80). Set Z based off A.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

LDX (AE)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Set X.L to Operand.L. Set N based off (X.L & \$80). Set Z based off X.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (AF)	Cycles: 5	Size: 4
Absolute Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Inc. PC .
4.2	Read Address:Bank	Store as Operand.
5.1		Set A.L to Operand.L. Set N based off (A.L & \$80). Set Z based off A.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

BCS (B0)	Cycles: 2	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch if C is 1.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

LDA (B1)	Cycles: 5	Size: 2
Direct Indirect Indexed (d),y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set A.L to Operand.L. Set N based off (A.L & \$80). Set Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (B2)	Cycles: 5	Size: 2
Direct Indirect (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer: DB	Store as Operand.
6.1		Set A.L to Operand.L. Set N based off (A.L & \$80). Set Z based off A.L .
6.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (B3)	Cycles: 7	Size: 2
Stack Relative Indirect Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set A .L to Operand.L. Set N based off (A .L & \$80). Set Z based off A .L.
7.2	Read PC:PB	Store as OpCode.
+X.1		

LDY (B4)	Cycles: 4	Size: 2
Direct, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Read PC:PB	Set Y.L to Operand.L. Set N based off (Y.L & \$80). Set Z based off Y.L .
5.2		Store as OpCode.
+X.1		

LDA (B5)	Cycles: 4	Size: 2
Direct, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Read PC:PB	Set A.L to Operand.L. Set N based off (A.L & \$80). Set Z based off A.L .
5.2		Store as OpCode.
+X.1		

LDX (B6)	Cycles: 4	Size: 2
Direct, Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + Y.L + D.L , Address.H = D.H , otherwise Address = Operand + Y + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Read PC:PB	Set X.L to Operand.L. Set N based off (X.L & \$80). Set Z based off X.L .
5.2		Store as OpCode.
+X.1		

LDA (B7) Cycles: 6		Size: 2
Direct Indirect Indexed Long [d],y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	If D.L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If D.L is not 0, read (Address + 2), otherwise read (Address & \$FF00) ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set A.L to Operand.L. Set N based off (A.L & \$80). Set Z based off A.L .
7.2	Read PC:PB	Store as OpCode.
+X.1		

CLV (B8)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set the V flag to 0.
2.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (B9)	Cycles: 4	Size: 3
Absolute, Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set A.L to Operand.L. Set N based off (A.L & \$80). Set Z based off A.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

TSX (BA)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform X.L = S.L , set N based off (X.L & \$80), set Z based off X.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

TYX (BB)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform X.L = Y.L , set N based off (X.L & \$80), set Z based off X.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

LDY (BC)	Cycles: 4	Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set Y .L to Operand.L. Set N based off (Y .L & \$80). Set Z based off Y .L.
5.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (BD)	Cycles: 4	Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set A.L to Operand.L. Set N based off (A.L & \$80). Set Z based off A.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

LDX (BE)	Cycles: 4	Size: 3
Absolute, Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set X.L to Operand.L. Set N based off (X.L & \$80). Set Z based off X.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

LDA (BF)	Cycles: 5	Size: 4
Absolute Long, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank += (Tmp >> 16).
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set A.L to Operand.L. Set N based off (A.L & \$80). Set Z based off A.L .
5.2	Read PC:PB	Store as OpCode.
+X.1		

CPY (C0)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC . Set C flag based off (Y .L >= Operand.L), set Z flag based off (Y .L == Operand.L), and set N flag based off ((Y .L - Operand.L) & \$80) != 0.
2.2		Store as OpCode.
+X.1		

CMP (C1) Cycles: 6 Size: 2		
Direct Indexed Indirect (d,x) (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Pointer to Pointer + X + D . Address.L = Pointer.L. Address.H = D .H.
2.2		If D .L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If D .L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer: DB	Store as Operand.
7.1		Set C flag based off (A .L >= Operand.L), set Z flag based off (A .L == Operand.L), and set N flag based off ((A .L - Operand.L) & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

REP (C2)	Cycles: 3	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		P &= ~(Operand.L & ~(X flag M flag)).
3.2		Store as OpCode.
+X.1		

CMP (C3)	Cycles: 4	Size: 2
Stack Relative (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1		Set C flag based off (A .L >= Operand.L), set Z flag based off (A .L == Operand.L), and set N flag based off ((A .L - Operand.L) & \$80) != 0.
4.2	Read PC:PB	Store as OpCode.
+X.1		

CPY (C4)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Set C flag based off (Y.L >= Operand.L), set Z flag based off (Y.L == Operand.L), and set N flag based off ((Y.L - Operand.L) & \$80) != 0.
4.2	Read PC:PB	Store as OpCode.
+X.1		

CMP (C5)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Set C flag based off (A.L >= Operand.L), set Z flag based off (A.L == Operand.L), and set N flag based off ((A.L - Operand.L) & \$80) != 0.
4.2	Read PC:PB	Store as OpCode.
+X.1		

DEC (C6)	Cycles: 5	Size: 2
Direct (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Perform Operand.L -= 1, set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

CMP (C7)		Cycles: 6	Size: 2
Direct Indirect Long (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	Read Address + 1	Store as Pointer.H.	
5.1			
5.2	Read Address + 2	Store as Bank.	
6.1			
6.2	Read Pointer:Bank	Store as Operand.	
7.1		Set C flag based off (A.L >= Operand.L), set Z flag based off (A.L == Operand.L), and set N flag based off ((A.L - Operand.L) & \$80) != 0.	
7.2	Read PC:PB	Store as OpCode.	
+X.1			

INY (C8)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform Y.L += 1, set N based off (Y.L & \$80), set Z based off Y.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

CMP (C9)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC . Set C flag based off (A.L >= Operand.L), set Z flag based off (A.L == Operand.L), and set N flag based off ((A.L - Operand.L) & \$80) != 0.
2.2		Store as OpCode.
+X.1		

DEX (CA)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform X.L -= 1, set N based off (X.L & \$80), set Z based off X.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

WAI (CB)	Cycles: 3	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Discard.
2.1		
2.2		
3.1		
3.2		Enter waiting state. Repeat this cycle forever.

CPY (CC)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Set C flag based off (Y .L >= Operand.L), set Z flag based off (Y .L == Operand.L), and set N flag based off ((Y .L - Operand.L) & \$80) != 0.
4.2	Read PC:PB	Store as OpCode.
+X.1		

CMP (CD)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Set C flag based off (A.L >= Operand.L), set Z flag based off (A.L == Operand.L), and set N flag based off ((A.L - Operand.L) & \$80) != 0.
4.2	Read PC:PB	Store as OpCode.
+X.1		

DEC (CE)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Perform Operand.L -= 1, set N based off (Operand.L & \$80) and Z based off Operand.L.
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

CMP (CF)	Cycles: 5	Size: 4
Absolute Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Inc. PC .
4.2	Read Address:Bank	Store as Operand.
5.1		Set C flag based off (A .L >= Operand.L), set Z flag based off (A .L == Operand.L), and set N flag based off ((A .L - Operand.L) & \$80) != 0.
5.2	Read PC:PB	Store as OpCode.
+X.1		

BNE (D0)	Cycles: 2	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch if Z is 0.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

CMP (D1) Cycles: 5		Size: 2
Direct Indirect Indexed (d),y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set C flag based off (A.L >= Operand.L), set Z flag based off (A.L == Operand.L), and set N flag based off ((A.L - Operand.L) & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

CMP (D2)		Cycles: 5	Size: 2
Direct Indirect (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Pointer.	
4.1			
4.2	Read Address + 1	Store as Pointer.H.	
5.1			
5.2	Read Pointer: DB	Store as Operand.	
6.1		Set C flag based off (A.L >= Operand.L), set Z flag based off (A.L == Operand.L), and set N flag based off ((A.L - Operand.L) & \$80) != 0.	
6.2	Read PC:PB	Store as OpCode.	
+X.1			

CMP (D3)	Cycles: 7	Size: 2
Stack Relative Indirect Indexed (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set C flag based off (A .L >= Operand.L), set Z flag based off (A .L == Operand.L), and set N flag based off ((A .L - Operand.L) & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

PEI (D4)	Cycles: 6	Size: 2
Direct Page Indirect		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		
4.2	If D.L is 0, Read (((Address.L + 1) & \$FF) (Address.H << 8)), otherwise read Address + 1	Store as Bank.
5.1		
5.2	Write to ((S.L \$0100)+0)	Push Operand.H onto stack.
6.1		
6.2	Write to ((S.L \$0100)-1)	Push Operand.L onto stack.
7.1		Dec. S.L by 2 and set S.H to 1.
7.2	Read PC:PB	Store as OpCode.
+X.1		

CMP (D5)	Cycles: 4	Size: 2
Direct, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		Set C flag based off (A.L >= Operand.L), set Z flag based off (A.L == Operand.L), and set N flag based off ((A.L - Operand.L) & \$80) != 0.
5.2	Read PC:PB	Store as OpCode.
+X.1		

DEC (D6)	Cycles: 6	Size: 2
Direct, X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Write to Address	Write Operand.L.
5.2		Perform Operand.L -= 1, set N based off (Operand.L & \$80), and Z based off Operand.L.
6.1		
6.2		Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

CMP (D7) Cycles: 6		Size: 2
Direct Indirect Indexed Long [d],y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	If D.L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If D.L is not 0, read (Address + 2), otherwise read (Address & \$FF00) ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.
7.1		Set C flag based off (A.L >= Operand.L), set Z flag based off (A.L == Operand.L), and set N flag based off ((A.L - Operand.L) & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

CLD (D8)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set the D flag to 0.
2.2	Read PC:PB	Store as OpCode.
+X.1		

CMP (D9) Cycles: 4		Size: 3
Absolute, Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set C flag based off (A .L >= Operand.L), set Z flag based off (A .L == Operand.L), and set N flag based off ((A .L - Operand.L) & \$80) != 0.
5.2	Read PC:PB	Store as OpCode.
+X.1		

PHX (DA)	Cycles: 3	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Write to ((S.L \$0100)+0)	Inc. PC .
1.2		
2.1		
2.2		Push X.L onto stack.
3.1		Dec. S.L by 1 and set S.H to 1.
3.2	Read PC:PB	Store as OpCode.
+X.1		

STP (DB)	Cycles: 4	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2		
2.1		
2.2		
3.1		
3.2		
4.1		Stop processor.

JML (DC)	Cycles: 6	Size: 3
Absolute Indirect Long (Jump)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC .
3.2	Read Pointer	Store as Operand.
4.1		
4.2	Read Pointer + 1	Store as Operand.H.
5.1		
5.2	If D.L is 0, Read (((Pointer.L + 2) & \$FF) (Pointer.H << 8)), otherwise read Pointer + 2	Store as Bank.
6.1		Set PC to Operand. Set PB to Bank.
6.2	Read PC:PB	Store as OpCode.
+X.1		

CMP (DD)		Cycles: 4	Size: 3
Absolute, X (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Pointer.	
2.1		Inc. PC .	
2.2	Read PC:PB	Store as Pointer.H.	
3.1		Inc. PC . Perform Orig = Pointer.	
		Tmp = ui32(Pointer + X).	
		Pointer = ui16(Tmp).	
		If Orig.H == Pointer.H, skip 2 half-cycles.	
		Bank = DB + (Tmp >> 16).	
3.2			
4.1			
4.2	Read Pointer:Bank	Store as Operand.	
5.1		Set C flag based off (A .L >= Operand.L), set Z flag based off (A .L == Operand.L), and set N flag based off ((A .L - Operand.L) & \$80) != 0.	
5.2	Read PC:PB	Store as OpCode.	
+X.1			

DEC (DE)	Cycles: 7	Size: 3
Absolute, X (Read/Modiy/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \mathbf{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\mathbf{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Perform $\text{Operand.L} -= 1$, set N based off ($\text{Operand.L} \& \$80$), and Z based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

CMP (DF) Cycles: 5 Size: 4		
Absolute Long, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Inc. PC . Perform Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
4.2	Read Pointer:Bank	Store as Operand.
5.1		Set C flag based off (A.L >= Operand.L), set Z flag based off (A.L == Operand.L), and set N flag based off ((A.L - Operand.L) & \$80) != 0.
5.2	Read PC:PB	Store as OpCode.
+X.1		

CPX (E0)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC . Set C flag based off (X .L >= Operand.L), set Z flag based off (X .L == Operand.L), and set N flag based off ((X .L - Operand.L) & \$80) != 0.
2.2		Store as OpCode.
+X.1		

SBC (E1) Cycles: 6		Size: 2
Direct Indexed Indirect (d,x) (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Pointer to Pointer + X + D . Address.L = Pointer.L. Address.H = D .H.
2.2		If D .L is 0, skip the next cycle.
3.1		Address.H = Pointer.H (fixes high byte of address).
3.2		
4.1		
4.2	Read Address	Store as Pointer.
5.1		
5.2	* If D .L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
6.1		
6.2	Read Pointer: DB	Store as Operand.
7.1		If D flag == 0: Result = ui16(A .L) - ui16(Operand) - (1 - C). V flag = ((ui16(A .L) ^ ui16(Operand)) & (ui16(A .L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A .L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A .L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A .L >> 4) ^ (Operand >> 4)) & ((A .L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

* The Single-Step Tests require this to be “Read Address + 1,” but this is an error in the tests.

SEP (E2)	Cycles: 3	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		
2.2		
3.1		Perform P = (Operand.L & ~(X flag M flag)).
3.2		Store as OpCode.
+X.1		

SBC (E3)	Cycles: 4	Size: 2
Stack Relative (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Operand.
4.1		If D flag == 0: Result = ui16(A .L) - ui16(Operand) - (1 - C). V flag = ((ui16(A .L) ^ ui16(Operand)) & (ui16(A .L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A .L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A .L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A .L >> 4) ^ (Operand >> 4)) & ((A .L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
4.2		Read PC:PB Store as OpCode.
+X.1		

CPX (E4)	Cycles: 3	Size: 2
Direct (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		Set C flag based off (X.L >= Operand.L), set Z flag based off (X.L == Operand.L), and set N flag based off ((X.L - Operand.L) & \$80) != 0.
4.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (E5)		Cycles: 3	Size: 2
Direct (Read)			
Cycle	R/W	Desc.	
-X.2	Read PC:PB	Store as OpCode.	
1.1		Inc. PC .	
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.	
2.1		Inc. PC .	
2.2			
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .	
3.2	Read Address	Store as Operand.	
4.1		If D flag == 0: Result = ui16(A.L) - ui16(Operand) - (1 - C). V flag = ((ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A.L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A.L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.	
4.2	Read PC:PB	Store as OpCode.	
+X.1			

INC (E6)	Cycles: 5	Size: 2
Direct (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Operand.
4.1		
4.2	Write to Address	Write Operand.L.
5.1		Perform Operand.L += 1, set N based off (Operand.L & \$80), set Z based off Operand.L.
5.2	Write to Address	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (E7) Cycles: 6		Size: 2
Direct Indirect Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Address + 2	Store as Bank.
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		If D flag == 0: Result = ui16(A.L) - ui16(Operand) - (1 - C). V flag = ((ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A.L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A.L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

INX (E8)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Perform X.L += 1, set N based off (X.L & \$80), set Z based off X.L .
2.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (E9)	Cycles: 2	Size: 2
Immediate		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC.
1.2	Read PC:PB	Store as Operand.
2.1		If D flag == 0: Result = ui16(A.L) - ui16(Operand) - (1 - C). V flag = ((ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A.L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A.L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
2.2	Read PC:PB	Store as OpCode.
+X.1		

NOP (EA)		Cycles: 2	Size: 1
Implied			
Cycle	R/W		Desc.
-X.2	Read PC:PB		Store as OpCode.
1.1			Inc. PC .
1.2	Read PC:PB		Discard.
2.1			
2.2	Read PC:PB		Store as OpCode.
+X.1			

XBA (EB)	Cycles: 3	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2		
2.1		
2.2		
3.1		Swap A.L and A.H , set N based off (A.L & \$80), and set Z based off A.L .
3.2	Read PC:PB	Store as OpCode.
+X.1		

CPX (EC)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		Set C flag based off (X .L >= Operand.L), set Z flag based off (X .L == Operand.L), and set N flag based off ((X .L - Operand.L) & \$80) != 0.
4.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (ED)	Cycles: 4	Size: 3
Absolute (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		If D flag == 0: Result = ui16(A.L) - ui16(Operand) - (1 - C). V flag = ((ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A.L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A.L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
4.2	Read PC:PB	Store as OpCode.
+X.1		

INC (EE)	Cycles: 6	Size: 3
Absolute (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read Address: DB	Store as Operand.
4.1		
4.2	Write to Address: DB	Write Operand.L.
5.1		Perform Operand.L += 1, set N based off (Operand.L & \$80), set Z based off Operand.L.
5.2	Write to Address: DB	Write Operand.L.
6.1		
6.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (EF)	Cycles: 5	Size: 4
Absolute Long (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Address.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Address.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Inc. PC .
4.2	Read Address:Bank	Store as Operand.
5.1		If D flag == 0: Result = ui16(A.L) - ui16(Operand) - (1 - C). V flag = ((ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A.L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A.L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
5.2	Read PC:PB	Store as OpCode.
+X.1		

BEQ (F0)	Cycles: 2	Size: 2
Relative		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC . Decide to branch if Z is 1.
1.2		Store as Operand. Address = i16(i8(Operand.L)) + PC , BoundaryCrossed = Address.H != PC .H. If not branching or not BoundaryCrossed, poll interrupts.
2.1		Inc. PC . If not branching, end (next half-cycle is 4.2).
2.2		
3.1		Set PC .L to Address.L. If not BoundaryCrossed, end (next half-cycle is 4.2).
3.2		Poll interrupts.
4.1		Set PC .H to Address.H
4.2		Store as OpCode.
+X.1		

SBC (F1) Cycles: 5		Size: 2
Direct Indirect Indexed (d),y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		If D flag == 0: Result = ui16(A.L) - ui16(Operand) - (1 - C). V flag = ((ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A.L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A.L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (F2) Cycles: 5		Size: 2
Direct Indirect (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		
5.2	Read Pointer: DB	Store as Operand.
6.1		If D flag == 0: $Result = ui16(A.L) - ui16(Operand) - (1 - C).$ $V\text{ flag} = ((ui16(A.L) \wedge ui16(Operand)) \& (ui16(A.L) \wedge ui8(Result)) \& \$80) \neq 0.$ $C\text{ flag} = Result \leq \$FF.$ $Result = ui8(Result).$ Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: $Lo = int16(A.L \& 0x0F) - int16(Operand \& 0x0F) - (1 - C);$ if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. $HiSum = int16(A.L \gg 4) - int16(Operand \gg 4) - BorrowToHi.$ $V\text{ flag} = (((A.L \gg 4) \wedge (Operand \gg 4)) \& ((A.L \gg 4) \wedge HiSum) \& \$08) \neq 0.$ if (HiSum < 0) HiSum -= 6. $C\text{ flag} = HiSum \geq 0.$ $Result = ui8(((HiSum \& 0x0F) \ll 4) \mid (Lo \& \$0F)).$ Z flag = Result == \$00. N flag = (Result & \$80) != 0.
6.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (F3) Cycles: 7		Size: 2
Stack Relative Indirect Indexed		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC . Set Address to Pointer + (S .L \$0100).
2.2		
3.1		
3.2	Read Address	Store as Pointer.
4.1		
4.2	Read Address + 1	Store as Pointer.H.
5.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank = DB + (Tmp >> 16).
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Operand.
7.1		If D flag == 0: Result = ui16(A .L) - ui16(Operand) - (1 - C). V flag = ((ui16(A .L) ^ ui16(Operand)) & (ui16(A .L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A .L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A .L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A .L >> 4) ^ (Operand >> 4)) & ((A .L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

PEA (F4)	Cycles: 5	Size: 3
Absolute (Push)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Operand.H.
3.1		Inc. PC .
3.2	Write to ((S .L \$0100)+0)	Push Operand.H onto stack.
4.1		
4.2	Write to ((S .L \$0100)-1)	Push Operand.L onto stack.
5.1		Dec. S .L by 2 and set S .H to 1.
5.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (F5) Cycles: 4		Size: 2
Direct, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1		If D flag == 0: Result = ui16(A.L) - ui16(Operand) - (1 - C). V flag = ((ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A.L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A.L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
5.2		Read PC:PB Store as OpCode.
+X.1		

INC (F6)	Cycles: 6	Size: 2
Direct, X (Read/Modify/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Store as Operand.
2.1		Inc. PC .
2.2		
3.1		If D.L is 0, Address.L = Operand + X.L + D.L , Address.H = D.H , otherwise Address = Operand + X + D . If D.L is 0, skip 2 half-cycles.
3.2		
4.1		
4.2	Read Address	Store as Operand.
5.1	Write to Address	Write Operand.L.
5.2		Perform Operand.L += 1, set N based off (Operand.L & \$80), set Z based off Operand.L.
6.1		Write Operand.L.
6.2		Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (F7)	Cycles: 6	Size: 2
Direct Indirect Indexed Long [d],y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Operand. If D.L is 0, skip the next cycle.
2.1		Inc. PC .
2.2		
3.1		Inc. PC if previous cycle was skipped. Set Address to Operand + D .
3.2	Read Address	Store as Pointer.
4.1		
4.2	If D.L is not 0, read (Address + 1), otherwise read (Address & \$FF00) ui8(Address.L + 1)	Store as Pointer.H.
5.1		
5.2	If D.L is not 0, read (Address + 2), otherwise read (Address & \$FF00) ui8(Address.L + 2)	Store as Bank.
6.1		Perform Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). Bank += (Tmp >> 16).
6.2	Read Pointer:Bank	Store as Operand.
7.1		If D flag == 0: Result = ui16(A.L) - ui16(Operand) - (1 - C). V flag = ((ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A.L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A.L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
7.2	Read PC:PB	Store as OpCode.
+X.1		

SED (F8)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Discard.
2.1		Set the D flag to 1.
2.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (F9) Cycles: 4		Size: 3
Absolute, Y (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC. Perform Orig = Pointer. Tmp = ui32(Pointer + Y). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		If D flag == 0: Result = ui16(A.L) - ui16(Operand) - (1 - C). V flag = ((ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A.L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A.L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
5.2	Read PC:PB	Store as OpCode.
+X.1		

PLX (FA)	Cycles: 4	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		
2.1		
2.2		
3.1		Inc. S.L by 1 and set S.H to 1.
3.2		Store as Operand.L.
4.1	Read u8(S.L +0) \$0100	Set X.L to Operand.L, set N based off (X.L & \$80), and set Z based off X.L .
4.2	Read PC:PB	Store as OpCode.
+X.1		

XCE (FB)	Cycles: 2	Size: 1
Implied		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1	Read PC:PB	Inc. PC .
1.2		Discard.
2.1		Exchanges the Carry and Emulation flags. If Emulation mode is entered, M and X flags are set to 1, S.H is set to 1, and X.H and Y.H are cleared to 0.
2.2		Store as OpCode.
+X.1		

JSR (FC)	Cycles: 8	Size: 3
Absolute Indexed Indirect (a,x) (Jump)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Write to u8(S .L+0) \$0100	Push PC .H onto stack.
3.1		
3.2	Write to u8(S .L-1) \$0100	Push PC .L onto stack.
4.1		Dec. S .L by 2 and set S .H to 1.
4.2	Read PC:PB	Store as Pointer.H.
5.1		Perform Pointer += X .
		Bank = PB .
5.2		
6.1		
6.2	Read Pointer:Bank	Store as Address.L.
7.1		
7.2	Read Pointer + 1:Bank	Store as Address.H.
8.1		Set PC to Address.
8.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (FD) Cycles: 4		Size: 3
Absolute, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform Orig = Pointer. Tmp = ui32(Pointer + X). Pointer = ui16(Tmp). If Orig.H == Pointer.H, skip 2 half-cycles. Bank = DB + (Tmp >> 16).
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		If D flag == 0: Result = ui16(A.L) - ui16(Operand) - (1 - C). V flag = ((ui16(A.L) ^ ui16(Operand)) & (ui16(A.L) ^ ui8(Result)) & \$80) != 0. C flag = Result <= \$FF. Result = ui8(Result). Z flag = Result == \$00. N flag = (Result & \$80) != 0. If D flag == 1: Lo = int16(A.L & 0x0F) - int16(Operand & 0x0F) - (1 - C); if (Lo < 0) Lo -= 6. BorrowToHi = Lo < 0 ? 1 : 0. HiSum = int16(A.L >> 4) - int16(Operand >> 4) - BorrowToHi. V flag = (((A.L >> 4) ^ (Operand >> 4)) & ((A.L >> 4) ^ HiSum) & \$08) != 0. if (HiSum < 0) HiSum -= 6. C flag = HiSum >= 0. Result = ui8(((HiSum & 0x0F) << 4) (Lo & \$0F)). Z flag = Result == \$00. N flag = (Result & \$80) != 0.
5.2	Read PC:PB	Store as OpCode.
+X.1		

INC (FE)	Cycles: 7	Size: 3
Absolute, X (Read/Modiy/Write)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$. Pointer = $\text{ui16}(\text{Tmp})$. Bank = $\text{DB} + (\text{Tmp} \gg 16)$.
3.2		
4.1		
4.2	Read Pointer:Bank	Store as Operand.
5.1		
5.2	Write to Pointer:Bank	Write Operand.L.
6.1		Perform $\text{Operand.L} += 1$, set N based off ($\text{Operand.L} \& \$80$), set Z based off Operand.L.
6.2	Write to Pointer:Bank	Write Operand.L.
7.1		
7.2	Read PC:PB	Store as OpCode.
+X.1		

SBC (FF)	Cycles: 5	Size: 4
Absolute Long, X (Read)		
Cycle	R/W	Desc.
-X.2	Read PC:PB	Store as OpCode.
1.1		Inc. PC .
1.2	Read PC:PB	Store as Pointer.
2.1		Inc. PC .
2.2	Read PC:PB	Store as Pointer.H.
3.1		Inc. PC .
3.2	Read PC:PB	Stores as Bank.
4.1		Inc. PC . Perform $\text{Tmp} = \text{ui32}(\text{Pointer} + \text{X})$. $\text{Pointer} = \text{ui16}(\text{Tmp})$. $\text{Bank} += (\text{Tmp} \gg 16)$.
4.2	Read Pointer:Bank	Store as Operand.
5.1		If D flag == 0: $\text{Result} = \text{ui16}(\text{A.L}) - \text{ui16}(\text{Operand}) - (1 - \text{C})$. V flag = $((\text{ui16}(\text{A.L}) \wedge \text{ui16}(\text{Operand})) \& (\text{ui16}(\text{A.L}) \wedge \text{ui8}(\text{Result}) \& \$80) \neq 0)$. C flag = $\text{Result} \leq \$\text{FF}$. $\text{Result} = \text{ui8}(\text{Result})$. Z flag = $\text{Result} == \$00$. N flag = $(\text{Result} \& \$80) \neq 0$. If D flag == 1: $\text{Lo} = \text{int16}(\text{A.L} \& 0x0F) - \text{int16}(\text{Operand} \& 0x0F) - (1 - \text{C})$; if $(\text{Lo} < 0)$ $\text{Lo} -= 6$. $\text{BorrowToHi} = \text{Lo} < 0 ? 1 : 0$. $\text{HiSum} = \text{int16}(\text{A.L} \gg 4) - \text{int16}(\text{Operand} \gg 4) - \text{BorrowToHi}$. V flag = $((\text{A.L} \gg 4) \wedge (\text{Operand} \gg 4)) \& ((\text{A.L} \gg 4) \wedge \text{HiSum}) \& \$08 \neq 0$. if $(\text{HiSum} < 0)$ $\text{HiSum} -= 6$. C flag = $\text{HiSum} \geq 0$. $\text{Result} = \text{ui8}(((\text{HiSum} \& 0x0F) \ll 4) \mid (\text{Lo} \& \$0F))$. Z flag = $\text{Result} == \$00$. N flag = $(\text{Result} \& \$80) \neq 0$.
5.2	Read PC:PB	Store as OpCode.
+X.1		